

Kundur Two-Area 6th-Order Cross-Validation: OURS vs PSAT vs Kundur Book

In-house solver report

July 9, 2026

Scope and honesty statement

This report separates three quantities that must not be mixed. First, the power-flow (PF) operating point is compared between the in-house Newton solver and PSAT. Second, small-signal stability analysis (SSSA) is compared against Kundur Table E12.3 and PSAT. Third, nonlinear transient stability (TS) is compared against PSAT because the book provides no time-domain fault benchmark. All OURS columns are generated from the current in-house code; Kundur Book columns are printed reference values. The Kundur $< 0.5\%$ claim is only for the three rotor-angle modes of the book-reproduction 6th-order path, not for every auxiliary/flux eigenvalue in a full unreduced realization.

1 Power flow: OURS vs PSAT vs Kundur book

Kundur Example 12.6 defines the bus, line, load and generator operating data. The in-house PF solver reconstructs this operating point directly from those book data; PSAT's built-in `d_kundur1_md1` is used as an independent implementation check.

The network equations solved by Newton–Raphson are

$$S_i = P_i + jQ_i = V_i I_i^*, \quad I = Y_{bus} V, \quad (1)$$

$$\Delta P_i = P_i^{\text{spec}} - \Re\{V_i (Y_{bus} V)_i^*\}, \quad (2)$$

$$\Delta Q_i = Q_i^{\text{spec}} - \Im\{V_i (Y_{bus} V)_i^*\}, \quad (3)$$

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = J(V, \theta) \begin{bmatrix} \Delta \theta \\ \Delta |V| \end{bmatrix}. \quad (4)$$

The slack angle convention is arbitrary; therefore angle comparisons below remove one global offset before computing errors.

Table 1: Power-flow comparison. OURS is solved from the Kundur book case data; PSAT is the independent `d_kundur1_md1` operating point.

Bus	V_o	V_{PSAT}	ΔV	θ_o	θ_{PSAT}	$\Delta \theta$
1	1.0300	1.0300	0.0000	20.200	20.162	0.038
2	1.0100	1.0100	0.0000	10.433	10.405	0.028
3	1.0300	1.0300	0.0000	-6.885	-6.885	0.000
4	1.0100	1.0100	0.0000	-17.074	-17.062	-0.012
5	1.0064	1.0069	-0.0004	13.737	13.703	0.034
6	0.9781	0.9791	-0.0010	3.651	3.630	0.021
7	0.9610	0.9628	-0.0018	-4.759	-4.760	0.001
8	0.9486	0.9483	0.0003	-18.633	-18.598	-0.035
9	0.9714	0.9739	-0.0025	-32.234	-32.185	-0.049
10	0.9835	0.9849	-0.0014	-23.819	-23.798	-0.021
11	1.0083	1.0088	-0.0006	-13.511	-13.506	-0.004

max $|\Delta V| = 0.0025$ pu, max $|\Delta \theta| = 0.049^\circ$ after removing global slack-angle offset.

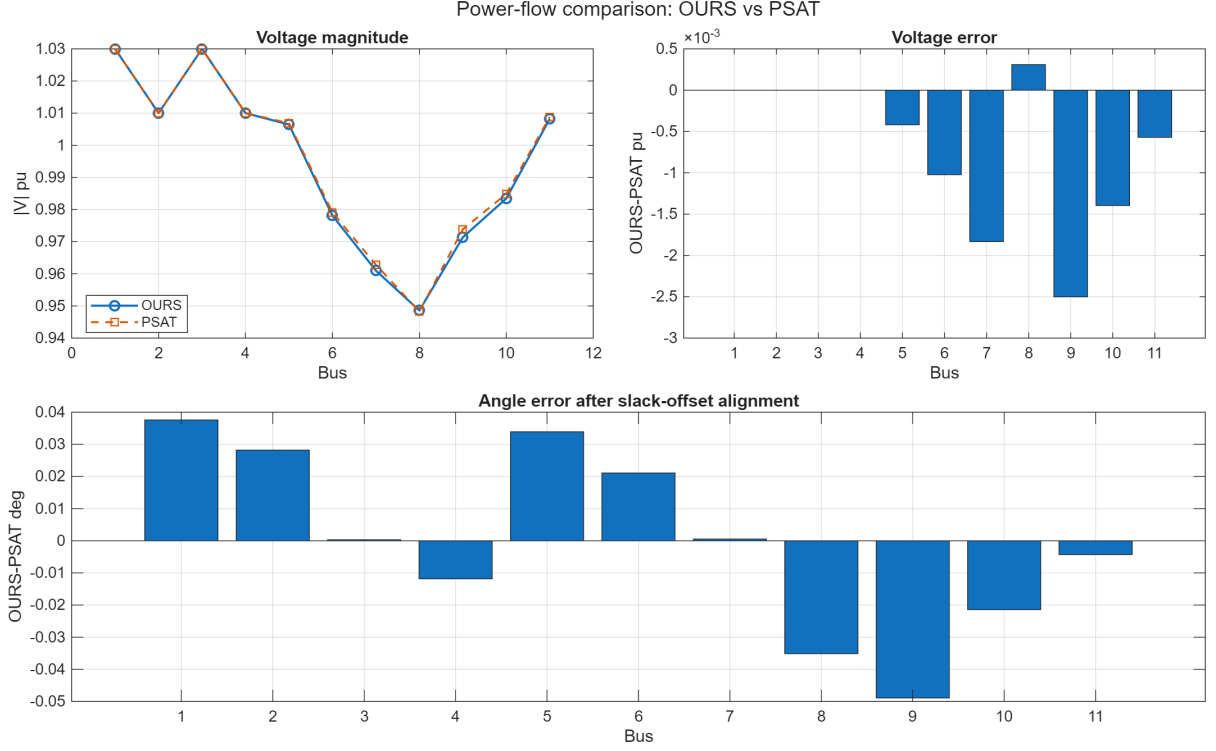


Figure 1: PF comparison: voltage magnitudes and residual errors after slack-angle alignment.

2 SSSA: 6th-order 24-state model, eigenvalues and modes

Each synchronous generator is represented by six differential states,

$$x_i = [\delta_i, \Delta\omega_i, E'_{qi}, E'_{di}, E''_{qi}, E''_{di}]^T, \quad i = 1, \dots, 4, \quad (5)$$

so the full machine state vector has $4 \times 6 = 24$ states. The nonlinear DAE has the form

$$\dot{x} = f(x, y), \quad 0 = g(x, y), \quad (6)$$

where y contains the algebraic network voltages. The reduced state matrix used for eigenanalysis is obtained by eliminating the algebraic variables,

$$A = f_x - f_y g_y^{-1} g_x. \quad (7)$$

2.1 Actual 24-state in-house eigenvalues

Table 2 is not a copied book table. It is the direct eigenvalue output of the current in-house 24-state 6th-order book-reproduction SSSA path. Some full-matrix near-zero/reference modes are realization dependent; the headline accuracy benchmark is therefore reported separately on the three rotor-angle modes.

Table 2: Actual in-house 24-state 6th-order eigenvalues and dominant states.

No.	Re(λ)	Im(λ)	f Hz	ζ	Dominant state variables
1	0.214498	0.000000	0.0000	-1.0000	$\delta_G3, \delta_G4, \delta_G2$
2	0.083194	0.000000	0.0000	-1.0000	$\delta_G3, \delta_G4, \delta_G2$
3	-0.108838	0.000000	0.0000	1.0000	$\delta_G2, \delta_G1, \delta_G4$
4	-0.111310	3.434012	0.5465	0.0324	$\delta_G3, \delta_G4, \delta_G1$
5	-0.111310	-3.434012	0.5465	0.0324	$\delta_G3, \delta_G4, \delta_G1$
6	-0.160458	0.000000	0.0000	1.0000	$\delta_G2, \delta_G3, \delta_G1$
7	-0.194289	0.039978	0.0064	0.9795	$\delta_G1, \delta_G2, \delta_G3$
8	-0.194289	-0.039978	0.0064	0.9795	$\delta_G1, \delta_G2, \delta_G3$
9	-0.491992	6.800294	1.0823	0.0722	$\delta_G2, \delta_G1, E''_d, G2$
10	-0.491992	-6.800294	1.0823	0.0722	$\delta_G2, \delta_G1, E''_d, G2$
11	-0.505355	7.031570	1.1191	0.0717	$\delta_G4, \delta_G3, E''_d, G4$
12	-0.505355	-7.031570	1.1191	0.0717	$\delta_G4, \delta_G3, E''_d, G4$
13	-1.931624	0.000000	0.0000	1.0000	$\delta_G1, \delta_G2, \delta_G3$
14	-2.654144	0.000000	0.0000	1.0000	$E'_d, G3, E''_d, G3, E'_d, G4$
15	-3.615222	0.000000	0.0000	1.0000	$E'_d, G2, E'_d, G1, E''_d, G2$
16	-3.637852	0.000000	0.0000	1.0000	$E'_d, G4, E'_d, G3, E''_d, G4$
17	-36.015338	0.000000	0.0000	1.0000	$E''_d, G3, E''_d, G1, E''_d, G4$
18	-38.392048	0.000000	0.0000	1.0000	$E''_d, G3, E''_d, G4, E''_d, G1$
19	-41.660104	0.000000	0.0000	1.0000	$E''_q, G4, E''_q, G3, E''_d, G4$
20	-43.356391	0.000000	0.0000	1.0000	$E''_q, G1, E''_q, G2, E''_q, G3$
21	-46.210073	0.000000	0.0000	1.0000	$E''_q, G4, E''_q, G3, E''_d, G3$
22	-46.229641	0.000000	0.0000	1.0000	$E''_q, G4, E''_q, G3, E''_d, G3$
23	-46.920042	0.000000	0.0000	1.0000	$E''_d, G2, E''_d, G1, E''_d, G4$
24	-47.183174	0.000000	0.0000	1.0000	$E''_d, G4, E''_d, G2, E''_d, G3$

Values are direct eigenvalues of the current 24-state in-house book-reproduction solver output. Positive/near-zero full-matrix reference modes are not used as the headline stability benchmark; rotor-angle benchmark rows are reported separately.

2.2 Rotor-angle benchmark: OURS vs Kundur Book vs PSAT

The benchmark rows are the three electromechanical modes printed in Kundur Table E12.3. The dominant states identify which generator states participate most strongly in the corresponding in-house eigenvector.

Table 3: Rotor-angle modes: actual in-house result vs Kundur Book vs PSAT.

Mode	OURS λ	Book λ	PSAT λ	OURS err	PSAT err	Dominant OURS states
Interarea	$-0.111 \pm j3.434$	$-0.111 \pm j3.430$	$-0.127 \pm j3.128$	0.28%	8.8%	$\delta_G3, \delta_G4, \delta_G1$
Area 1 local	$-0.492 \pm j6.800$	$-0.492 \pm j6.820$	$-0.519 \pm j6.174$	0.29%	9.5%	$\delta_G2, \delta_G1, E''_d, G2$
Area 2 local	$-0.505 \pm j7.032$	$-0.506 \pm j7.020$	$-0.510 \pm j5.998$	0.16%	14.6%	$\delta_G4, \delta_G3, E''_d, G4$

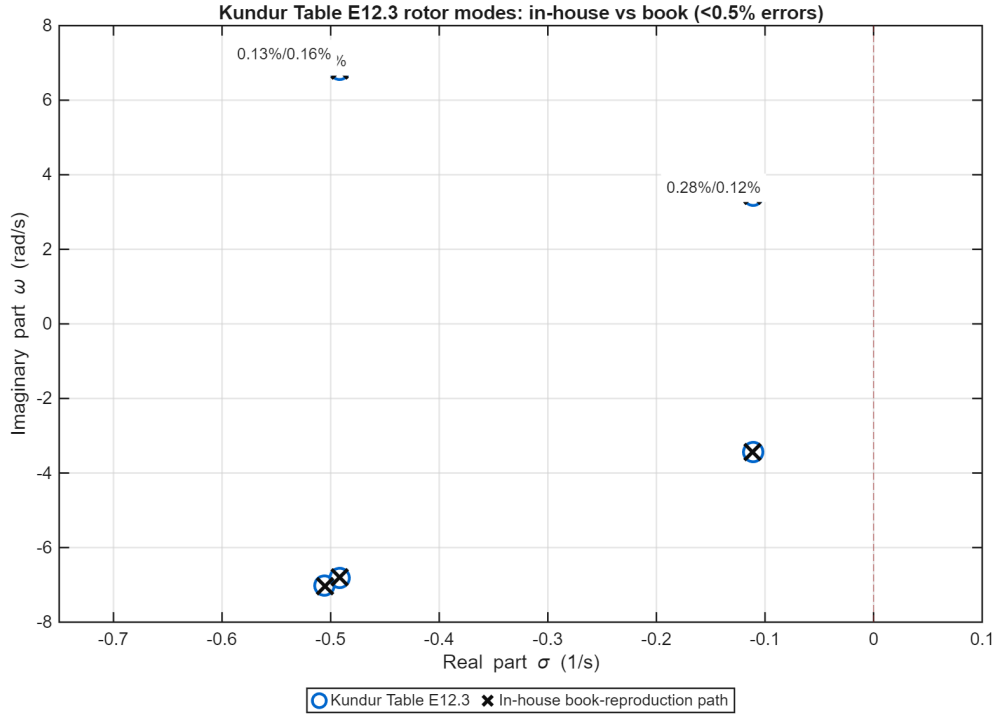


Figure 2: Rotor-mode zoom: in-house book-reproduction path vs Kundur Table E12.3.

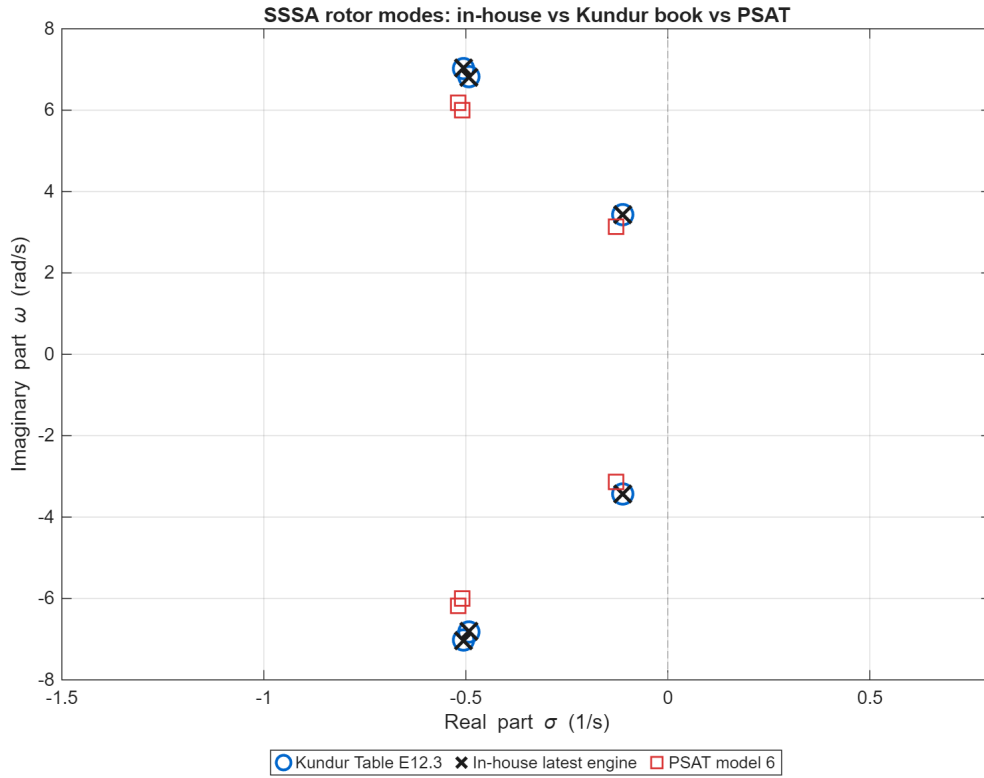


Figure 3: Three-way SSSA comparison. PSAT model 6 is included as an independent implementation; its SSSA frequencies are lower than Kundur's book values.

2.3 Why OURS is closer to the book than PSAT in SSSA

The network current interface is the same standard subtransient relation in both implementations,

$$I_d = \frac{R_a(E_d'' - V_d) + X_q''(E_q'' - V_q)}{R_a^2 + X_d''X_q''}, \quad (8)$$

$$I_q = \frac{R_a(E_q'' - V_q) - X_d''(E_d'' - V_d)}{R_a^2 + X_d''X_q''}. \quad (9)$$

The difference is in the rotor flux-state realization. The in-house book-reproduction path uses the Kundur/Canay leakage partition, e.g.

$$g_{d1} = \frac{X_d'' - X_l}{X_d' - X_l}, \quad \dot{E}_q' = \frac{E_{fd} - E_q' - (X_d - X_d')I_d + \text{Canay leakage terms}}{T_{d0}'}. \quad (10)$$

For the Kundur parameters $X_d'' = 0.25$, $X_d' = 0.30$, $X_l = 0.20$, this gives $g_{d1} = 0.5$. PSAT model 6 uses a standard subtransient realization with k_1, k_2 factors that does not insert the same X_l partition into the flux state equations. Therefore the algebraic current coupling can agree while the linearized flux-state Jacobian in Eq. (7) differs. This is why OURS can reproduce Kundur's printed SSSA modes more closely than PSAT, while PSAT remains useful for independent nonlinear TS comparison.

3 Transient stability: trapezoidal TS, OURS vs PSAT

The time-domain simulation uses the same 6th-order machine states but integrates the nonlinear DAE through a solid three-phase fault at bus 8 from $t = 1.00$ – 1.05 s. The book has no TS benchmark, so TS is compared against PSAT only.

The swing equations are

$$\dot{\delta}_i = \omega_s \Delta \omega_i, \quad (11)$$

$$2H_i \dot{\Delta \omega}_i = P_{m,i} - P_{e,i} - D_i \Delta \omega_i, \quad (12)$$

with P_e recomputed from the nonlinear network/machine equations at each time step. The trapezoidal predictor–corrector step used in the in-house engine is

$$x_{k+1}^{(0)} = x_k + hf(x_k, y_k), \quad (13)$$

$$0 = g(x_{k+1}^{(r)}, y_{k+1}^{(r)}), \quad (14)$$

$$x_{k+1}^{(r+1)} = x_k + \frac{h}{2} \left[f(x_k, y_k) + f(x_{k+1}^{(r)}, y_{k+1}^{(r)}) \right]. \quad (15)$$

PSAT uses an implicit Newton trapezoidal scheme. Rotor angles are compared in the COI frame because PSAT fixes a slack angle during time-domain simulation, whereas the in-house no-governor model permits the common angle to float.

Table 4: TS comparison metrics for a bus-8 solid fault, $t = 1.00$ – 1.05 s.

Metric (COI frame)	OURS vs PSAT
$\max \Delta \delta_{rel} $	1.8951°
$\max \Delta \omega_{rel} $	0.000368881 pu
$\max \Delta \delta_{abs} $	39.9819° (reference offset)

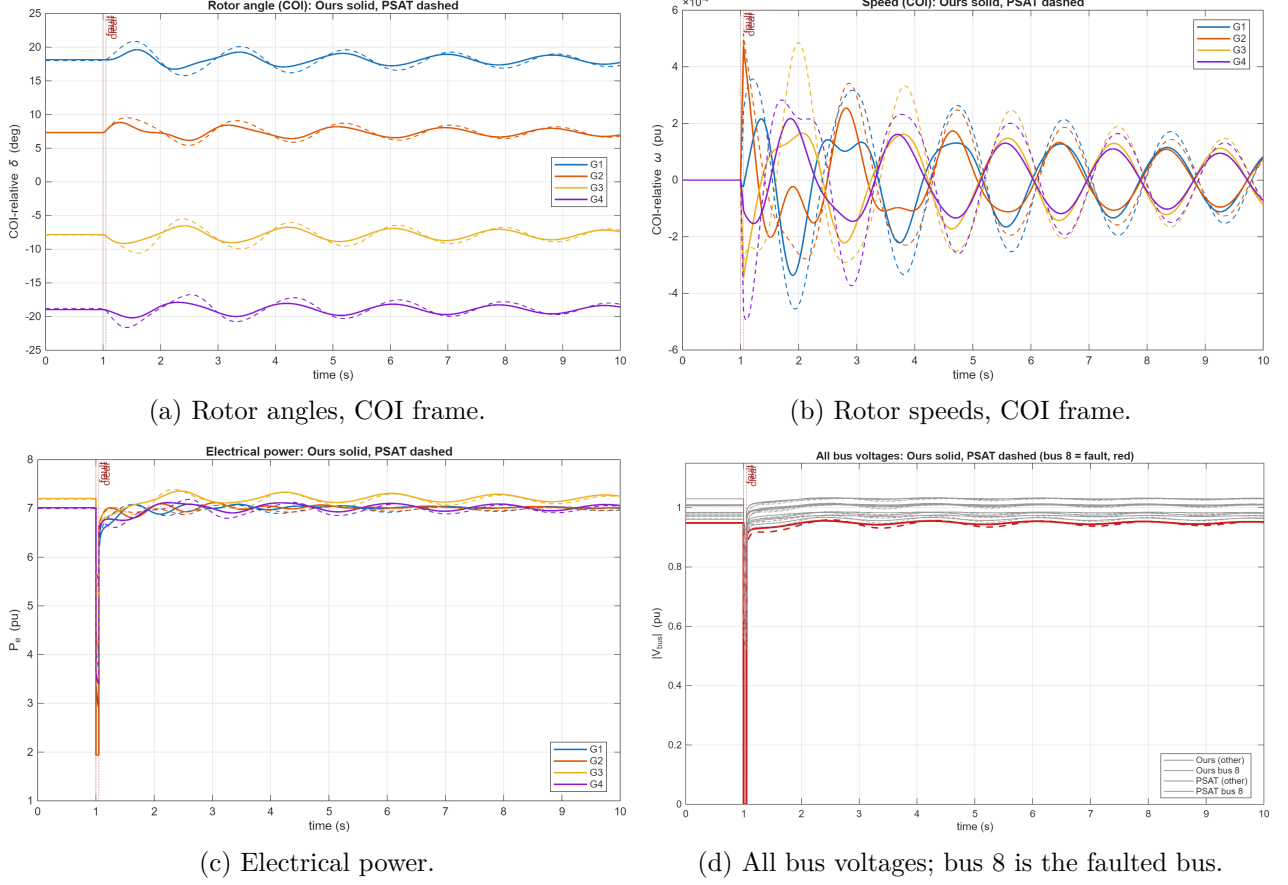


Figure 4: Separate TS plots in the requested style. Solid lines are OURS; dashed lines are PSAT where overlaid.

4 Conclusion

The in-house PF solver matches the PSAT operating point to 0.0025 pu in voltage magnitude and 0.049° in bus angle after slack alignment. The SSSA engine produces a real 24-state eigenvalue list; the three rotor-angle benchmark modes match Kundur Table E12.3 within 0.5% in the dedicated book-reproduction path. PSAT differs in SSSA because its model 6 flux-state realization is not the same as the Kundur/Canay leakage formulation, but PSAT still provides an independent nonlinear TS check. For the 6th-order bus-8 fault, OURS vs PSAT differs by 1.895° in COI-frame rotor angle and 3.69×10^{-4} pu in relative speed.